

Contact

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Top Skills

Neuromorphic Engineering
Applied Machine Learning
Data Analysis

Languages

Arabic
Hebrew
English

Certifications

edX Honor Code Certificate for
Introduction to Computer Science
and Programming Using Python

Honors-Awards

Best poster award
Magna cum laude
Schulich Leader Scholarships - for
excellent students in STEM areas

Ben Efron

Postdoctoral Researcher | Brain-Inspired Computing, Neuromorphic
Engineering & AI | Neuroscience & Data Science
Leuven, Flemish Region, Belgium

Summary

I am a neuroscientist and postdoctoral researcher at the University of Liège, specializing in brain-inspired computing and neuromorphic engineering. My work bridges neuroscience, AI, and robotics to develop adaptive algorithms and intelligent systems.

With over 7 years of experience in systems neuroscience, electrophysiology, and large-scale data analysis, I excel at translating complex biological signals into actionable insights and machine intelligence solutions. I design and run experiments, acquire and synchronize multi-modal datasets, and implement data-driven models that inform both neuroscience and AI applications.

My research focuses on applying principles from biological sensory systems to neuromorphic computing, robotics, and AI, creating solutions that are both efficient and adaptive. I enjoy collaborating across disciplines to turn fundamental neuroscience insights into innovative, real-world technologies.

I am open to connecting with professionals and organizations exploring AI, neuromorphic computing, robotics, and data-driven innovations.

Experience

University of Liège
Postdoctoral Researcher
July 2024 - Present (1 year 4 months)
Liège, Walloon Region, Belgium

Postdoctoral research in the Brain-Inspired Computing Lab.

- Lead experimental design and execution in the biological branch of the Brain-Inspired Computing Lab.
- Develop neuromorphic algorithms inspired by sensory systems and validate them in robotic and computational frameworks.

- Acquire, synchronize, and analyze multi-modal neuronal datasets using HD-MEAs and advanced data processing pipelines.
- Integrate neuroscience principles with machine learning and AI to develop adaptive and efficient intelligent systems.
- Collaborate with electrical engineering and life sciences teams to bridge biology, robotics, and AI research.

Skills: Neuromorphic Engineering | Data Analysis | Python | MATLAB | Machine Learning | Electrophysiology | Algorithm Development | Experimental Design | Robotics

Weizmann Institute of Science

7 years 3 months

PHD Graduate Student

April 2017 - June 2024 (7 years 3 months)

Rehovot Area, Israel

Apr 2017 – Jun 2024 | Israel

- Investigated auditory information processing in the mouse cortex during natural behavior using electrophysiology and behavioral experiments.
- Designed and executed multi-sensor experiments, acquiring and analyzing large datasets from neuronal populations and instruments.
- Applied statistical modeling and machine learning techniques to interpret complex neural signals.
- Developed computational models informed by neuroscience to explore sensory integration and network dynamics.

MSc Student

April 2017 - March 2020 (3 years)

Rehovot, Central, Israel

Ben Efron- lighting

Lighting Technician

March 2017 - October 2017 (8 months)

Ramot Meir, HaMerkaz (Central) District, Israel

Self employed lighting technician and lighting programmer, head of lighting staffs, head of technical staffs.

Davidson institute of science education

Guide

March 2017 - April 2017 (2 months)

Rehovot, HaMerkaz (Central) District, Israel

Guide in the garden of science, open air science museum.

Jerusalem Lions Rugby

Team Manager

January 2006 - April 2017 (11 years 4 months)

Player, captain and team manager of the Jerusalem rugby team. In charge of social media promotion, advertisement and Hebrew university cooperation.

Florey Institute of Neuroscience and Mental Health

Research Assistant

July 2016 - December 2016 (6 months)

Melbourne, Victoria, Australia

Building an air track system to simulate a maze environment for head-fixed mice.

Beit Avi Chai

Member Of Technical Staff

March 2014 - May 2016 (2 years 3 months)

Jerusalem, Israel

A/V, Sound, and lighting engineering, backstage, lighting operator.

Magenta sound and light

Lighting Engineer

April 2010 - May 2016 (6 years 2 months)

Netiv HaLamed-Heh, Jerusalem District, Israel

Light and sound technician and stage manager. Lighting programmer and designer. Head of on site team. Dance, music, theater, art shows, festivals etc.

Prof. Hermona Soreq laboratory

B.A. student research

September 2014 - March 2016 (1 year 7 months)

Jerusalem, Israel

miR regulation and gene expression analysis via bioinformatics tools.

Miki Golan

Trail safety constructor

March 2011 - November 2012 (1 year 9 months)

Building and maintaining trekking and hiking trails and safety facilities in them.

Education

Weizmann Institute of Science

Master of Science - MS, Systems Neuroscience · (October 2017 - November 2019)

The Hebrew University

Bachelor's Degree, Physiological Psychology/Psychobiology · (2013 - 2016)

University of Melbourne

Exchange program, Physiological Psychology/Psychobiology · (2016 - 2016)